LIGHTYEAR

MS 100 Public COBOL execution and scenario calibration

Reconciled October 14 2026.

This milestone provides an executable engineering baseline before customer mainframe access. Original public INTCALC and POSTTRAN programs run through a pinned GnuCOBOL toolchain. Instrumented scenarios show which decisions were observed and whether deliberately changed COBOL behavior was detected. These measurements help prepare a customer engagement; they do not replace z/OS evidence.

Delivered capability

Source hashes, copybooks, compiler and runtime identities, generated file adapters, clocks and output hashes are bound in build and run receipts. Two clean hosted builds produced identical hashes for all ten executables. The public three-way INTCALC comparison runs the twin, Python reference and Java candidate on the same inputs and retains disagreements.

Decision instrumentation counts observed branch outcomes. Deterministic public scenarios add boundary, rounding and cross-record cases. The early generated campaign increased observed outcomes from 51 to 64 out of 94 for INTCALC, and from 63 to 73 out of 110 for POSTTRAN. Its eight hand-picked mutants are a small experiment, not a systematic adequacy estimate.

Later systematic records distinguish executed differences, execution failures, survivors, compiler refusals and evidence gaps. The retained October 12 report records 308 raw kills out of 421 INTCALC sites and 360 out of 483 POSTTRAN sites. Nine previously misclassified POSTTRAN execution refusals receive no kill credit. Scoped exclusions and individual acceptance are separate from those raw counts.

Generic scenario foundation and remaining work

The new generic engine first step translates hash-bound fixed-format source and explicitly supplied copybooks into typed expressions, assignments, conditions and conservative dependency edges. PIC and USAGE layouts, numeric domains and source locations are retained. Unsupported constructs produce located issues. This is not a solver or a program-equivalence proof; steps two through four have not been implemented by this delivery.

The retained systematic report still has missing valid targeted comparisons, insufficient field diversity and review conditions. Platform I/O, indexed ordering, native VSAM, CICS, Db2 and z/OS services remain outside the twin claim. Passing repository CI does not activate a release threshold or erase a historical evidence gap.

Supporting records

- Executable twin and hosted build evidence
- Three-way reconciliation
- Measured scenario generation
- Retained systematic counts and exact experiment identity
- Generic scenario step one